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Set 5: Website Analytics

Dataset: Website Traffic

Date Page Views Click-through Rate

1. Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.
2. Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.
3. Develop a stacked area chart to display the distribution of user interactions (likes, shares, comments) on a website.
4. Build a Tableau dashboard with the line chart and bar chart for interactive exploration of website traffic data.

SOLUTION :

CODE :

```
import pandas as pd
import matplotlib.pyplot as plt
import plotly.express as px
import plotly.graph_objects as gg
from plotly.subplots import make_subplots

df_traffic = pd.DataFrame({
    'Date': ['2023-01-01', '2023-01-02', '2023-01-03', '2023-01-04', '2023-01-05'],
    'Page_Views': [1500, 1600, 1400, 1650, 1800],
    'CTR': [2.3, 2.7, 2.0, 2.4, 2.6]
})
```

```
df_interactions = pd.DataFrame({
    'Date': ['2023-01-01', '2023-01-02', '2023-01-03', '2023-01-04', '2023-01-05'],
    'Likes': [120, 150, 100, 140, 180],
    'Shares': [40, 55, 30, 45, 60],
    'Comments': [15, 25, 10, 20, 30]
})
```

```
plt.figure(figsize=(8, 4))
plt.plot(df_traffic['Date'], df_traffic['Page_Views'], marker='o', color='purple', linewidth=2)
plt.title('Daily Page Views Trend', fontsize=14)
plt.xlabel('Date', fontsize=12)
plt.ylabel('Page Views', fontsize=12)
plt.xticks(rotation=15)
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
```

N = 3

```
top_ctr = df_traffic.sort_values(by='CTR', ascending=False).head(N)
```

```
plt.figure(figsize=(8, 4))
plt.bar(top_ctr['Date'], top_ctr['CTR'], color='darkorange', edgecolor='black')
plt.title(f'Top {N} Days with Highest Click-Through Rates (CTR)', fontsize=14)
plt.xlabel('Date', fontsize=12)
plt.ylabel('Click-Through Rate (%)', fontsize=12)
for i, val in enumerate(top_ctr['CTR']):
    plt.text(i, val + 0.05, f'{val}%', ha='center', fontweight='bold')
plt.tight_layout()
```

```

plt.show()

plt.figure(figsize=(8, 4))
plt.stackplot(
    df_interactions['Date'],
    df_interactions['Likes'],
    df_interactions['Shares'],
    df_interactions['Comments'],
    labels=['Likes', 'Shares', 'Comments'],
    colors=['#6baed6', '#74c476', '#fdae6b'],
    alpha=0.85
)
plt.title('Distribution of User Interactions Over Time', fontsize=14)
plt.xlabel('Date', fontsize=12)
plt.ylabel('Interaction Count', fontsize=12)
plt.legend(loc='upper left')
plt.xticks(rotation=15)
plt.tight_layout()
plt.show()

fig = make_subplots(
    rows=1, cols=2,
    subplot_titles=("Daily Page Views", "Top CTR Days")
)

fig.add_trace(gg.Scatter(x=df_traffic['Date'], y=df_traffic['Page_Views'],
mode='lines+markers', name='Page Views'), row=1, col=1)

fig.add_trace(gg.Bar(x=top_ctr['Date'], y=top_ctr['CTR'], marker_color='orange',
name='CTR (%)'), row=1, col=2)

```

```
fig.update_layout(title_text="Set 5: Website Analytics Interactive Dashboard", height=450)
fig.show()
```

OUTPUT :



